Juliette Bruce's Future Plans

My research interests lie in algebraic geometry and commutative algebra, which are two fields that study the same objects from opposite sides. A parabola in the plane can be described geometrically as the set of points equidistant from a fixed point and a fixed line, or algebraically as the solutions to the equation $y = x^2$. Each of these descriptions provides a different perspective on the same questions. For example, the fact that a straight line intersects a parabola in at most two points (geometry) reflects the fact that a quadratic equation has at most two solutions (algebra). This dictionary between algebra and geometry is at the heart of my field and motivates much of my research. Questions about shapes (lines, parabolas, curves, surfaces, etc.) become questions about polynomial equations and vice versa.

A running theme throughout much of my research is that the algebra of the polynomial equations defining a shape captures its geometry in subtle ways. For example, the equations defining a shape are rarely independent. Instead, the equations normally satisfy a separate set of relations among themselves, called syzygies (from the Greek for "yoked together"). The equations defining these relations then satisfy their own relations, and so on. Amazingly, Hilbert showed that this cascade of relations eventually terminates, resulting in a finite sequence of relations (i.e., syzygies) that captures aspects of the geometry of our shape.

Over the next several years, my research program seeks to carry this dictionary into new settings in three closely connected directions. The first two directions seek to extend the classical story of syzygies and a measure of geometric complexity called Castelnuovo–Mumford regularity to settings that have long been out of reach: curves with extra symmetries (stacky curves) and shapes built out of polyhedra (toric varieties). The third direction seeks to add a third language to the algebra–geometry dictionary discussed above: combinatorics. Each directions seeks to tabkle longstanding, hard geometric questions by translating them into algebra and combinatorics, which are more amenable to explicit calculation.

Syzygies of Stacky Curves. Curves are among the most classical objects studied in algebraic geometry and mathematics more broadly. A forty-year-old program begun by Mark Green, and advanced by breakthrough work of Claire Voisin, studies how the syzygies of a curve in its most natural embedding detect how special the curve is. The program's central conjecture, known as Green's conjecture for canonical curves, is one of the jewels of the field, and has motivated significant work and generalizations.

Stacky curves are curves where we encode extra symmetries by allowing a finite number of special points on the curve, called stacky points, to labeled with groups of symmetries. Such spaces are not exotic and arise naturally. For example, the spaces that organize modular forms, highly symmetric functions at the heart of number theory and central to the proof of Fermat's Last Theorem, are stacky curves. Despite this prevalence, the classical theory of syzygies does not directly apply to them, as they require variables whose degrees are not all one. The equations of stacky curves were only recently described by Voight and Zureick-Brown, and their syzygies remain uncharted.

With M. Banks and P. Wei, I am building this missing theory. We have already found the shape of the syzygies for stacky canonical curves with only a few special stacky points. This has led us to make a series of conjectures that generalize Green's conjecture to stacky canonical curves: the special points shift the classical answer in a predictable way (by exactly the number of extra equations they require). In the next several years, I hope to develop the tools needed to prove these conjectures. Additionally, these conjectures point towards a much larger and richer theory of syzygies of stacky curves in general.

Multigraded Regularity. Classically, the Castelnuovo–Mumford regularity of an algebraic variety is a single number that measures the complexity of the shape. It turns out to be central to many questions about defining equations and computation in algebraic geometry. This importance is underlined by a celebrated 1984 theorem of Eisenbud and Goto showing that seemingly different geometric and algebraic measures of complexity give the same answer: Castelnuovo–Mumford regularity.

However, many important shapes, such as the stacky curves in the previous section, do not live in ordinary projective space, but instead in spaces built by gluing affine pieces together in ways determined by polyhedra. For these more general ambient spaces, Maclagan and Smith introduced a generalization of Castelnuovo–Mumford regularity called multigraded regularity. In this theory, regularity is no longer a single number, but instead is an infinite region of lattice points bounded by a staircase-shaped boundary that can be remarkably intricate. For instance, with my collaborators and I showed that even if one is only

looking at the simplest possible geometric shape, a point in a toric surface, the staircase can have infinitely many corners, a pathology with no classical analog.

My plans aim to understand and tame this behavior. First, I have conjectures that the infinite staircases are an artifact of measuring complexity with respect to a cone that does not capture all of the geometry. In particular, I aim to show that there is a precise duality between the regularity staircase and the set of curves on a toric variety. If successful, this would not only improve our theoretical understanding of multigraded regularity, but also provide the first algorithm for computing multigraded regularity in full. Second, I aim to prove a multigraded analog of a theorem of Gruson, Lazarsfeld, and Peskine that bounds the Castelnuovo–Mumford regularity of a curve in terms of its degree—the number of points where it meets a general hyperplane (for example, a parabola has degree 2). The difficulty of such a generalization is finding the right multigraded analog of degree for curves on toric varieties, since the multigraded regularity we wish to bound is not a number but an infinite staircase.

Finally, and more speculatively, I will explore whether regularity is asking the right question, but in the wrong algebraic setting. Recent work has shown that the algebra better captures the geometry of toric varieties when one studies a toric variety and all of its close relatives at once. Inspired by this, I suspect a similar phenomenon holds for multigraded regularity. In particular, I would like to give a new definition of regularity using a recently introduced algebraic tool called the Cox category. If successful, this theory would resolve Oda's conjecture, a longstanding open problem in toric and polyhedral geometry.

Tropical Geometry of Moduli Spaces. A *moduli space* is a space whose points themselves represent shapes, with nearby points representing similar shapes. Many basic questions about the topology of such spaces remain unanswered. For example, Grushevsky asked whether the moduli space of abelian varieties has nonzero odd-degree cohomology classes, which detect holes in odd dimensions.Tropical geometry offers a way in: it replaces complicated objects with their combinatorial shadows. Curves are replaced with graphs, abelian varieties are represented by polyhedral data, and so on. Building on breakthrough work of Chan, Galatius, and Payne for curves, my collaborators and I adapted these methods to abelian varieties. This provided numerous new cohomology classes and answered Grushevsky's question affirmatively. Since this moduli space is closely related to groups of integer matrices, our results also give new information about these groups.

A central aspect of my research program over the next several years is to develop these methods into general tools. Work of Deligne shows that the cohomology of a moduli space has a natural filtration whose layers are indexed by weights. The work of Chan–Galatius–Payne and of my collaborators and me shows how tropical geometry captures one of these layers, namely the weight-zero layer of compactly supported cohomology, which is entirely determined by the combinatorial shadow of the moduli space. My plan for the future is to push this further in three directions. First, I aim to enrich the tropical spaces we study so that they themselves are moduli spaces. This requires using a broader notion of space, called a cone stack. Together with Raluca Vlad, I have proved preliminary results suggesting that this approach will be successful. Second, I hope to give a precise geometric comparison showing how the weight-zero layer can be described explicitly from the tropical moduli space. Third, I will push into the deeper, more mysterious layers of cohomology by studying the weight-two compactly supported cohomology of the moduli space of abelian varieties. Doing this requires enriching the classical combinatorics of tropical geometry so that the combinatorial shadows carry slightly more geometric information.

Computation, Open-Source Software, & Community-Centered AI. Computing examples to develop conjectures and survey the landscape of a problem is a central driver of much of my theoretical work. For example, using high-performance, high-throughput computing, my collaborators and I computed the syzygies of hundreds of new surfaces beyond previous reach (*Experimental Mathematics / Journal of Algebra*). Further, I have co-authored four software packages for the open-source computer algebra system *Macaulay2*. I plan to continue building upon this work in three ways. First, since no software exists for computing with stacky curves, M. Banks, J. Cobb, and I are developing `toricStacks`, a package for *Macaulay2*. Second, I am building a searchable database of thousands of expert-chosen mathematical objects and examples (ideals, rings, modules, etc.) that are hidden in the source code and documentation of *Macaulay2*. By making these examples easily accessible, this database will make testing conjectures, benchmarking software, and developing community-centered AI easier. Third, with J. Voight, I am planning a workshop focused on the future of mathematical software in the age of AI.